Uniform Local Tabularity in Intuitionistic Logic

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Abstract

By contrast wih S4, the analysis of local tabularity above IPC has provided a difficult challenge. This paper studies a strengthening of local tabularity – *uniform local tabularity* – where one demands that all formulas be equivalent to formulas of a given implication depth. Algebraically, this amounts to considering Heyting algebras generated by finitely many iterations of the implication operation. It is shown that in contrast with locally finite Heyting algebras, n-uniformly locally finite Heyting algebras always form a variety, and an explicit axiomatization of the variety of n-uniform locally finite Heyting algebras for $n \leqslant 3$ is given. In connection with this analysis, it is shown that there exist locally tabular logics which are not uniformly locally tabular, answering a question of Shehtman – an example of a pre-uniformly locally tabular logic is presented, which is shown to be the unique pre-uniformly locally tabular extension of the system KG.

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1 Introduction

In intuitionistic and modal logics, one often looks for properties which ensure the system is "sufficiently simple" to analyze in depth. Local tabularity ensures this by saying that over finitely many propositional variables there can only be finitely many such formulas, and in turn, that the semantics of the system cannot

be too complicated. This appears as a strengthening of 'tabularity' – saying that a single finite structure captures the full semantics of the logic – which is quite well-understood in intuitionistic and modal logics (see [12, Chapter 12]). Moreover, it is a classical result due to Segerberg [25] and Maksimova [20] that an extension of S4 is locally tabular if and only if it is of finite depth. This provides an easy criterion and algorithm to verify local tabularity, with there being a single "pre-locally tabular" logic [12, Sec. 12.4].

By contrast, in the system IPC, the situation is quite different. It was shown by Mardaev [22, 21] that there is a continuum of pre-locally tabular logics, all above KC. It is open whether every locally tabular logic is contained in a pre-locally tabular one and whether local tabularity is decidable. In terms of algebraic models – which correspond in algebraic logical terms to the logics – it is thus open whether the category of locally finite Heyting algebra admits any easy and logically insightful description (see [19] for recent discussion).

Given these difficulties it is reasonable to seek for strengthenings of local tabularity. In [28], Shehtman observed that one can use bisimulation games to characterize when certain logics are locally tabular, introducing a property called "finite formula depth", which corresponds to the existence of a bound on the nestings of modal operators or implications. In [2], we dubbed this property "uniform local tabularity", with the idea being that the bound on modal depth or implication depth provides a uniform bound on the size of formulas. It was shown by Shehtman that above S4 local tabularity implies uniform local tabularity, via finite depth, and hence the 'niceness' of local tabularity may be attributed to the uniform version. The relationship between local tabularity and its uniform version was likewise noted in [26], where generalizations of the finite height criterion were discussed. The concept of uniform local tabularity also has deep connections to the theory of free Heyting algebras and free constructions of Heyting algebras: as discussed in [3, Chapter 5], uniformity of local tabularity appears to be connected to the existence of simple representations of duals of free algebras.

This article provides a first systematic investigation of this property in the context of superintuitionistic logics. Our main contributions are the following:

1. We provide an explicit axiomatization of the varieties of n-uniform Heyting algebras for $n \leqslant 3$; for $n = 1$, this corresponds to the logic CPC, and for $n = 2$, the extensions of logic LC, the Gödel-Dummett calculus [13, 18]; for $n = 3$, such a logic is introduced here, and called 3Uni, being the least 3-uniform logic, and sitting between well-known logics:

$$\mathsf{wPL} \subsetneq \mathsf{3Uni} \subsetneq \mathsf{LC},$$

   where wPL is the logic of the Weak Peirce's Law (see Section 3.1).

2. We provide an example of a locally tabular logic which is not uniformly locally tabular, answering a problem left open in [27]. Indeed, we show that the logic Box (see Section 4), is pre-uniformly locally tabular – all of its extensions are uniformly locally tabular, though Box itself is not so. We also show that it is the unique logic with this property above the system KG [1].

2 Preliminaries and basic theory

2.1 Superintuitionistic logics and Kripke models

Throughout we will assume that we are working in the signature $\mathcal{L} = (\wedge, \vee, \rightarrow, \bot, \top)$ of Heyting algebras, the language of intuitionistic logic. We use the letters $\varphi, \psi, \ldots$ for $\mathcal{L}$-formulas and letters $p, q, \ldots$ or $x, y, \ldots$ for variables; we compactly represent a tuple of distinct variables as $\overline{p}$. We write $\varphi(\overline{p})$ to mean that all the variables of φ occur amongst $\overline{p}$.

Definition 2.1. The *implication-depth* of φ, denoted by $d(\varphi)$, is defined by induction:

- $d(p) = d(\bot) = d(\top) = 0$;
- $d(\varphi \wedge \psi) = d(\varphi \vee \psi) = \max(d(\varphi), d(\psi))$;

- $d(\varphi \to \psi) = \max(d(\varphi), d(\psi)) + 1$.

Given a poset P, we write $P \Vdash \varphi$ to mean that P validates φ in the usual Kripke semantics (see e.g., [12, Chapter 3]). We denote by Ext(IPC) the lattice of its extensions; elements are called *superintuitionistic logics*. Given a poset P, we denote by:

$$\mathrm{Log}(P) = \{\varphi : P \Vdash \varphi\},$$

the *logic of* P, which is an element of Ext(IPC). If $P \Vdash L$ for all $\varphi \in L$ we say that P *is a frame for* L.

Given a poset P, a subset $U \subseteq P$ is called an *upset* if whenever $x \in U$ and $x \leqslant y$ then $y \in U$. We denote the set of upsets of P by $\mathsf{Up}(P)$. Given a subset C, we denote by $\uparrow C$ the least upset containing it, and for $x \in P$, $\uparrow x := \uparrow\{x\}$. We say that Q is a *rooted upset* of P if there is $x \in P$ such that $Q = \uparrow x$. Given $x \in P$, we also denote $\uparrow x$ by P_x.

Definition 2.2. A *p-morphism* between posets $f : P \to Q$ is an order-preserving function such that for each $x \in P$ and $y \in Q$ if $f(x) \leqslant y$ then there is some $x' \geqslant x$ such that $f(x') = y$; we say that Q is a *p-morphic image* of P if there is a surjective p-morphism from P onto Q.

We now recall some relevant notions concerning finite Kripke models; the reader can find all necessary background in [17, Chapter 4.1].

Definition 2.3. We say that a triple $(\mathfrak{M}, x) = (P, \leqslant, c, x)$ where $(P, \leqslant)$ is a rooted poset, with root x, and $c : P \to \mathcal{P}(\overline{p})$ is an order-preserving function, is a *model over the propositional letters* $\overline{p}$. We say that $(\mathfrak{M}, x)$ is a model of a logic L (over the propositional letters $\overline{p}$) if $(P, \leqslant)$ is a frame for L. Moreover, given $\mathbf{K}$ a class of posets, we say that $(\mathfrak{M}, x)$ a model over $\overline{p}$ is a *$\overline{p}$-model of* $\mathbf{K}$ if there is some poset $Q \in \mathbf{K}$ such that P is a p-morphic image of a rooted upset of Q. We denote by $\mathcal{M}_{\overline{p}}(\mathbf{K})$ the class of $\overline{p}$-generated models over $\mathbf{K}$. We also denote by $M_{\mathbf{K}}(\varphi) = \{(\mathfrak{M}, x) \in \mathcal{M}_{\overline{p}}(\mathbf{K}) : (\mathfrak{M}, x) \Vdash \varphi\}$.

Definition 2.4. Given two models $(\mathfrak{M}, x) = (P, \leqslant, c, x)$ and $(\mathfrak{N}, y) = (Q, \leqslant, c', y)$ over $\overline{p}$, we say that they are *bisimilar* if there exists a relation $S \subseteq P \times Q$ such that (1) $(x, y) \in S$; (2) Whenever wSw' then $c(w) = c'(w')$; (3) whenever wSw' and $w \leqslant v$ there is some $w' \leqslant v'$ such that vSv'; (4) whenever wSw' and $w' \leqslant v'$ there is some $w \leqslant v'$ such that vSv'. We say that two points $w \in P$ and $w' \in Q$ are bisimilar if (P_w, w) is bisimilar with $(Q_{w'}, w')$.

Remark 2.1.

1. (**Notation**) Whenever we do not wish to emphasize the specific propositional letters used in the language,we will identify the set $\{p_1, \ldots, p_n\}$ with the natural number n, and consider models over $\overline{p}$ to be maps $c : P \to 2^n$; the image for a point $x \in P$ will be called its *color*, denoted by a tuple $c(x) = \langle i_1, \ldots, i_n \rangle$. We will further shorten tuples of the form $\langle 1, \ldots, 1, 0, \ldots, 0 \rangle$, consisting of a sequence of 1 followed by a sequence of 0, by the notation $1^{n-k}0^k$. When n is given, we will often omit the order and the valuation, and write simply (P, x) for a model. We occasionally write $\mathcal{M}_n(\mathbf{K})$ for the class of models over n from $\mathbf{K}$.

2. All statements given below for posets (P, x) or models thereof assume that P is *rooted at* x. This is implicit in our notion of model, but we repeat it for the sake of the reader familiar with frame theoretic arguments.

3. Given an n, say that a model (P, x) over n is *n-generated* if P does not contain distinct bisimilar points. A well-known characterization of n-generation which we will use throughout is through α and β-reductions (see [8, Chapter 3]). Namely, given a model P, we say that we can apply an *alpha-reduction* (α) whenever $x \in P$ has an immediate successor x' such that $\uparrow x = (\uparrow x' \cup \{x\})$ and $c(x) = c(x')$, obtained by identifying x with x'; we say we can perform a β-reduction (β) if x, x' are two points, such that $\uparrow x - \{x\} = \uparrow x' - \{x'\}$, and $c(x) = c(x')$, obtained by identifying x and x'. A model is n-generated precisely when there are no α or β-reductions which can be performed on the model. A convenient fact which we will exploit throughout is that for n-generated models, *bisimulation implies isomorphism*.

Definition 2.5. Let $(P, \leqslant, c, x)$ and $(Q, \leqslant, c', y)$ be two models over n. We say that a relation $S_k \subseteq P \times Q$ is a *k-bisimulation* based on (x, y) if there are relations $S_k \subseteq \cdots \subseteq S_j \subseteq \cdots \subseteq S_0 \subseteq P \times Q$ for each $0 \leqslant j \leqslant k$, such that xS_ky and for each $w \in P, w' \in Q$:

1. Whenever wS_0w' then $c(w) = c'(w')$;
2. For each $j < k$, whenever $wS_{j+1}w'$ and $w \leqslant v$ there is some $w' \leqslant v'$ such that vS_jv'.
3. For each $j < k$, whenever $wS_{j+1}w'$ and $w' \leqslant v'$ there is some $w \leqslant v$ such that vS_jv'.

We say that (P, x) is k-bisimilar with (Q, y) if there exists a k-bisimulation between them. We write $(P, x) \leqslant_0 (Q, y)$ if $c(x) \leqslant c(y)$ and $(P, x) \leqslant_{k+1} (Q, y)$ if for each $y \leqslant w'$ there is some $x \leqslant w$ such that $(P_w, w) \sim_k (Q_{w'}, w')$. We also write $(P, x) \leqslant_\infty (Q, y)$ if whenever $y \leqslant w'$ there is some $x \leqslant w$ such that (P, w) is bisimilar with (W, w')

As described in [17, Chapter 4.1], n-bisimulations can be characterized in game-theoretic terms: given a pair (P, x) and (Q, y), Player I wins a 0-bisimulation game if $c(x) \neq c(y)$ and otherwise Player II wins; and Player I wins an $n+1$-bisimulation game if there is $z \in P$, such that Player II cannot choose any $w \in Q$ and win an n-bisimulation game with (P_z, z) and (Q_w, w).

The following is a basic property of n-bisimulation:

Proposition 2.6. For each pair of models (P, x) and (Q, y) over $\overline{p}$ the following are equivalent:

1. $(P, x) \leqslant_n (Q, y)$;
2. Whenever $d(\varphi(\overline{p})) \leqslant n$ and $(P, x) \Vdash \varphi$ then $(Q, y) \Vdash \varphi$.

It follows from this that for finite models, $(P, x) \leqslant_\infty (Q, y)$ if and only if for every formula φ, $(P, x) \Vdash \varphi$ then $(Q, y) \Vdash \varphi$. The following is completely analogous to [17, Chapter 4, Exercise 11], where we relativise to a specific subclass of posets, to capture different equivalence in different logics:

Proposition 2.7. Let $\mathbf{K}$ be a class of finite posets, $L = \mathrm{Log}(\mathbf{K})$ and let $\overline{p}$ be a set of propositional variables. Then we have:

1. For each $\varphi(\overline{p}), \psi(\overline{p})$, we have:
$$L \vdash \varphi \leftrightarrow \psi \iff \mathcal{M}_{\mathbf{K}}(\varphi) = \mathcal{M}_{\mathbf{K}}(\psi).$$
2. The following are equivalent:
   (a) $S \subseteq \mathcal{M}_{\overline{p}}(\mathbf{K})$ is the class of models from $\mathcal{M}_{\overline{p}}(\mathbf{K})$ satisfying a formula ψ with $d(\psi) \leqslant n$;
   (b) S is closed under $\leqslant_n$-bisimulation within $\mathcal{M}_{\overline{p}}(\mathbf{K})$: whenever $(P, x) \in S$ and and $(Q, y) \in \mathcal{M}_{\overline{p}}(\mathbf{K})$ and $(P, x) \leqslant_n (Q, y)$ then $(Q, y) \in S$

Proof. (1) is an immediate consequence of L being the logic of a class of finite posets, and therefore, if $\varphi \leftrightarrow \psi \notin L$, there being a model of φ which is not a model of ψ.

(2) If S is the class of models satisfying a formula ψ with $d(\psi) \leqslant n$, then by Proposition 2.6, we have that S is closed under $\leqslant_n$-bisimulation. Conversely, assume that S is closed under $\leqslant_n$-bisimulation, and consider:
$$\mathsf{Th}_n(S) = \{\psi(\overline{p}) : d(\psi) \leqslant n, \forall (P, x) \in S, (P, x) \Vdash \psi\}.$$
Since there are only finitely many formulas of implication depth at most n, $\mathsf{Th}(S)$ is finite, and so we can consider $\chi(S) = \bigwedge \mathsf{Th}(S)$. We claim that S is precisely the class of models from $\mathcal{M}_{\overline{p}}(\mathbf{K})$ satisfying $\chi(S)$. It is clear that if $(P, x) \in S$, $(P, x) \Vdash \chi(S)$. On the other hand, assume that $(Q, y) \notin S$. Then for each $(P, x) \in S$, $(P, x) \nleqslant_n (Q, y)$, which by Proposition 2.6 means that there is a formula $\varphi(\overline{p})$ of implication depth at most n such that $(P, x) \Vdash \varphi_P$ and $(Q, y) \nVdash \varphi_P$. Let $\mu_P = \bigvee_{P \in S} \varphi_P$; note that up to equivalence, μ_P is finite, and $\mu_P \in \mathsf{Th}_n(S)$. Since $(Q, y) \nVdash \mu_P$, then $(Q, y) \nVdash \chi(S)$, which concludes the proof. ■

2.2 Local tabularity, local finiteness and their uniform versions

To investigate these notions it will be convenient to work with Heyting algebras. For the connection between superintuitionistic logics and varieties of Heyting algebras, and the algebraic semantics therein, we refer to [12, Chapter 8]. Moreover, though the reader will not explicitly need any facts about the duality of finite Heyting algebras and finite posets, this will be implicit in the following results.

Given a formula φ, we write $H \vDash \varphi$ to mean that the Heyting algebra H validates φ; notations are extended to classes in the usual way. Given a class of Heyting algebras $\mathbf{K}$, we write:

$$\mathrm{Log}(\mathbf{K}) = \{\varphi : \mathbf{K} \vDash \varphi\};$$

conversely, given a logic L, we write:

$$\mathrm{Alg}(L) = \{H \in \mathbf{HA} : H \vDash L\}.$$

We now turn to the basic concepts in our analysis.

Definition 2.8. Let $L \supseteq \mathsf{IPC}$ be a superintuitionistic logic. We say that L is:

- *locally tabular* if given $\overline{p} = \{p_1, \ldots, p_n\}$ propositional variables, there are only finitely many formulas over $\overline{p}$ up to L-provability. We denote by LTab the class of locally tabular logics;
- *n-uniform* if each formula φ is equivalent to a formula of implication depth n over L. We say that L is *uniformly locally tabular* if L is n-uniform for some n. We denote by ULTab_n the class of n-uniform logics, and by ULTab their union, the class of all uniformly locally tabular logics;
- of *finite depth* if $L \supseteq \mathsf{BD}_n$ for some n, where:

  $$\mathsf{bd}_0 = p_0 \text{ and } \mathsf{bd}_{n+1} = p_{n+1} \vee (p_{n+1} \to \mathsf{bd}_n),$$

  and

  $$\mathsf{BD}_n = \mathsf{IPC} \oplus \mathsf{bd}_n.$$

  We denote by FinDep the class of logics of finite depth.
- *tabular* if there is a finite poset P such that $L = \mathrm{Log}(P)$. We write Tab for the class of tabular logics.

All of these concepts correspond algebraically to specific properties:

Definition 2.9. Let H be an algebra. We say that H is *locally finite* if whenever $A \leqslant H$ is a finitely generated algebra, then A is finite. We say that a class of algebras $\mathbf{K}$ is *locally finite* if every algebra in $\mathbf{K}$ is locally finite.

Recall that given H a Heyting algebra, $A \subseteq H$ a subset, the least Heyting subalgebra of H containing A can be constructed as:

$$\langle A \rangle = \{t(b_1, \ldots, b_k) : b_i = a_i \text{ or } b_i = \neg a_i, a_i \in A,\ t(x_1, \ldots, x_n) \text{ is a formula }\}.$$

Local finiteness means precisely that for each H, and A a finite subset, there is an n such that $\langle A \rangle$ is generated by formulas of implication depth at most n. This is where we can impose uniformity:

Definition 2.10. Let H be an algebra. We say that H is *n-uniformly finite*, or simply *n-uniform*, if whenever $A \subseteq H$ is finite, then:

$$\langle A \rangle = \{t(a_1, \ldots, a_k) : a_i \in A, t(x_1, \ldots, x_k) \text{ has implication depth } \leqslant n\}.$$

In this case we say that $\langle A \rangle$ is *n-uniformly generated*. We say that a class of algebras $\mathbf{K}$ is *n-uniformly* finite if every algebra in $\mathbf{K}$ is n-uniformly finite.

The motivation for the former definition lies in a certain modal understanding of Heyting algebras: one sees the implications as modalities and their nesting as a way to parametrize the complexity of a given logical system or algebra. A different way of measuring such complexity goes through the structure of frames:

Definition 2.11. Given $x \in P$ a poset, the *depth of* x, denoted $d_p(x)$, is the largest n such that there exists a chain $x = x_1 \leqslant \cdots \leqslant x_n$, and there is no such chain of size $n+1$, or ∞ if such an n does not exist. If each point in P has a depth, we denote by $d_p(P) = \sup\{d_p(x) : x \in P\}$.

The following is all well-known and provides the algebraic translation of the above properties:

Proposition 2.12. The following holds for L a superintuitionistic logic:

1. L is locally tabular if and only if $\mathsf{Alg}(L)$ is locally finite.
2. L is n-uniform if and only if $\mathsf{Alg}(L)$ is n-uniformly finite.
3. L is of finite depth k if and only if whenever $P \Vdash L$ is a poset, then P is of depth at most k, if and only if L contains the formula bd_k, defined recursively by:

$$\mathsf{bd}_1 = p_1 \vee \neg p_1 \text{ and } \mathsf{bd}_{j+1} = p_{j+1} \vee (p_{j+1} \to \mathsf{bd}_j)$$

4. L is tabular if and only if $\mathsf{Alg}(L)$ is finitely generated as a variety (i.e., it is the least variety containing a given finite algebra).

Moreover, the following inclusions hold between classes of Heyting algebras:

$$\mathsf{Tab} \subseteq \mathsf{FinDep} \subseteq \mathsf{ULTab} \subseteq \mathsf{LTab}.$$

Proof. The equivalences of (1-3) are easy to prove. We concentrate on the inclusions. $\mathsf{Tab} \subseteq \mathsf{FDep}$, since each finite poset P satisfies bd_k where k is its height; the fact that $\mathsf{FDep} \subseteq \mathsf{ULTab}$ was shown by Shehtman [27], and amounts to the fact that for posets of finite depth, there is a uniform bound for how complex n-bisimulation can be; the fact that $\mathsf{ULTab} \subseteq \mathsf{LTab}$ is obvious. ■

Most converses of the inclusions of Proposition 2.12 are known not to hold. Namely, the logic BD_2 is known not to be tabular, even though it is of finite depth. The logic LC is axiomatized as:

$$\mathsf{LC} := (p \to q) \vee (q \to p),$$

is known to be uniformly locally tabular (see [27], see also [2] for a proof) but is not of finite depth. We will take a look at the last converse – $\mathsf{LTab} \subseteq \mathsf{ULTab}$ in Section 3.

2.3 Basic properties and criteria for uniform local tabularity

In order to establish n-uniform local tabularity we will make use of the following characterization in terms of n-bisimulations[1]:

Proposition 2.13. Let $\mathbf{K}$ be a class of finite posets. Then $L = \mathsf{Log}(\mathbf{K})$ is $n+1$-uniformly locally tabular if and only if for each $\overline{p}$, whenever (P, x) and (Q, y) are two models in $\mathcal{M}_{\overline{p}}(\mathbf{K})$ and $(P, x) \leqslant_{n+1} (Q, y)$, then $(P, x) \leqslant_\infty (Q, y)$. Moreover, if n-bisimilarity implies full bisimilarity, the latter holds.

[1] We note for the reader that the criterion below makes use of the relation $\leqslant_n$ as opposed to n-bisimilarity. In [27], the corresponding result does not mention this relation, however, we have not been able to reproduce such a result when omitting this relation.

Proof. If L is $n+1$-uniformly locally tabular, then all formulas are equivalent to formulas of implication depth $\leqslant n+1$. Let $(P,x),(Q,y) \in \mathcal{M}_{\overline{p}}(\mathbf{K})$ be two models, and assume that $(P,x) \leqslant_{n+1} (Q,y)$. Then whenever $(P,x) \Vdash \varphi$ for any φ, $(Q,y) \Vdash \varphi$. Thus $(P,x) \leqslant_{\infty} (Q,y)$ by our remark after Proposition 2.6.

Conversely, let $\varphi(\overline{p})$ be an arbitrary formula. Consider $S = \{(P,x) \in \mathcal{M}_{\overline{p}}(\mathbf{K}) : (P,x) \Vdash \varphi\}$. Now assume that $(P,x) \leqslant_{n+1} (Q,y)$. By assumption, $(P,x) \leqslant_{\infty} (Q,y)$, and consequently, $(Q,y) \Vdash \varphi$ as well. This means that means that S is closed under $\leqslant_{n+1}$-bisimulation within $\mathcal{M}_{\overline{p}}(\mathbf{K})$, which by Proposition 2.7.(2) that S is the class of models from $\mathcal{M}_{\overline{p}}(\mathbf{K})$ satisfying a formula ψ with $d(\psi) \leqslant n+1$. By Proposition 2.7.(1), this means that $L \vdash \varphi \leftrightarrow \psi$, i.e., φ is equivalent to a formula of implication depth $\leqslant n+1$. Thus L is $n+1$-uniformly locally tabular, as desired.

If we assume that n-bisimilarity implies full bisimilarity instead, then if we assume that $(P,x) \leqslant_{n+1} (Q,y)$, by considering $y \leqslant y$, we have that there is some $x \leqslant w$ such that $(P_w,w) \sim_n (Q,y)$. Since $(P,x) \Vdash \varphi$, then $(P_w,w) \Vdash \varphi$ as well; since n-bisimilarity implies full bisimilarity, by invariance of truth under bisimilarity, $(Q,y) \Vdash \varphi$. We thus obtain the same conclusion. ■

We close this section by noting an interesting property of n-uniform Heyting algebras. Let $\mathbf{HA}_{lf}$ be the category of locally finite Heyting algebras with Heyting algebra homomorphisms, and let $\mathbf{HA}^n_{lf}$ be the category of n-uniformly finite Heyting algebras with the same maps.

Proposition 2.14. The category $\mathbf{HA}_{lf}$ is closed under subalgebras, homomorphic images and finite products. For each $n \in \omega$, $\mathbf{HA}^n_{lf}$ is a variety.

Proof. The fact about subalgebras is trivial. To see the fact about homomorphic images, say H is locally finite (n-uniformly finite), and A is a homomorphic image. Let $\{b_1,\dots,b_n\} \subseteq A$ be finitely many elements. Then we can find finitely many elements $\{h_1,\dots,h_n\} \subseteq H$, mapping surjectively onto b_1; by restricting the homomorphism from H to A, we find that $\langle\{h_1,\dots,h_n\}\rangle$ maps surjectively to $\langle\{b_1\dots,b_n\}\rangle$. Thus the latter is finite (n-uniformly generated) by hypothesis.

To see the fact about products, note that if H_0, H_1 are locally finite, and $(a_0,b_0),\dots,(a_n,b_n)$ where $a_i \in H_0$ and $b_i \in H_1$, then

$$\langle(a_0,b_0),\dots,(a_n,b_n)\rangle = \{t((a_0,b_0),\dots,(a_n,b_n)) : d(t) \text{ is the maximum of that of } \langle A\rangle \text{ and } \langle B\rangle.$$

The same argument works for infinite products if we assume the algebras are n-uniformly finite, since then, rather than taking the maximum of $\langle A\rangle$ and $\langle B\rangle$ one can take the bound n. ■

The above Proposition entails that there is the least n-uniform logic for each n. On the other hand, $\mathbf{HA}_{lf}$ is not a variety – it contains every finite Heyting algebra, and hence the smallest variety containing it is $\mathbf{HA}$ itself.

3 The hierarchy of locally tabular logics

In this section we will study the hierarchy of locally tabular logics – the *top* of the Kuznetsov hierarchy [7]. In 3.1 by characterizing the $0,1,2,3$-uniformly locally tabular logics. In Section 3.2 we make some remarks on 3-uniform logics of infinite depth, connecting them to recent studies of local tabularity in bi-intuitionistic logic [24]. We then give in Section 3.3 an example of a locally tabular but not uniformly locally tabular logic.

3.1 0-uniform, 1-uniform, 2-uniform, 3-uniform logics

The reader will notice that due to our definition, the only superintuitionistic logic which is 0-uniform is the trivial logic. 1-uniformity turns out to be quite a restrictive property: only CPC has it. 2-uniformity will be characteristic of extensions of LC, whilst 3-uniformity and beyond will turn out to have a rather complex structure.

We assume the reader is familiar with the Yankov theorem (see [8, Chapter 3] for an extensive discussion) associating a formula $\mathcal{J}(Q)$ to a rooted poset Q; the key property we will need is the following:

Theorem 3.1 (Yankov's theorem)**.** *For a finite poset P and a rooted poset Q, $P \nVdash \mathcal{J}(Q)$ if and only if Q is a rooted subframe of a p-morphic image of P.*

We briefly recall the structure of the lattice of extensions of superintuitionistic logics; all proofs of these facts can be found in [12, Chapter 4, Chapter 8]:

1. Given logics $L' \subseteq L$ we say that L' is a *cover* of L if L is an immediate successor of L', i.e., $L' \subseteq K \subseteq L$ implies that either $L' = K$ or $K = L$.
2. CPC is the largest consistent extension of IPC;
3. CPC has a unique cover $\mathsf{Sme} = \mathsf{IPC} \oplus (\neg p \vee q \vee (q \to p)) = \mathsf{Log}(2)$, the logic of the 2-element poset.
4. Sme is an extension of LC, in fact the greatest extension different from CPC.
5. The logic LC is axiomatized by $\mathcal{J}(\mathsf{2F})$ and $\mathcal{J}(\mathsf{Dia})$ where $\mathsf{2F}$ is the 2-fork (the frame underlying the model M_2 in Figure 1), and Dia is the diamond frame (the poset Q_6 in Figure 2).

Then we have:

Theorem 3.2. *For L a superintuitionistic logic, we have:*

1. *L is 0-uniform if and only if L is inconsistent.*
2. *L is 1-uniform if and only if $L = \mathsf{CPC}$;*
3. *L is 2-uniform if and only if $L \supseteq \mathsf{LC}$.*

Proof. As noted, the only 0-uniform logic is the inconsistent one. Note that CPC is 1-uniform: by Definition 2.1, every formula in CPC is equivalent to a conjunction of disjunctions of formulas $p, \neg p$. On the other hand, the logic Sme is not 1-uniform: we see in Figure 1 models M_0 and M_1, based on frames validating Sme, and $M_1 \leqslant_1 M_0$, certainly, but $M_1 \nleqslant_\infty M_0$, since no point in M_1 labelled with 0 is 1-bisimilar with the singleton containing 0. By Proposition 2.13, this means that the logic is not 1-uniform, and neither is any logic extended by it.

As noted before, it was shown by Shehtman [27] that LC is 2-uniform; it is also not difficult to see: if (P, x) and (Q, y) are two linearly ordered posets rooted at x, y, they are n-generated, and they are 1-bisimilar, then notice that by n-generation, there cannot be successive nodes with the same color (otherwise we could identify them). But then by an easy induction we see that the two chains $\uparrow x$ and $\uparrow y$ must be isomorphic. It follows that if $(P, x) \sim_1 (Q, y)$ for n-generated models, then (P, x) and (Q, y) are isomorphic, and consequently, fully bisimilar. We can then appeal to the same proposition as above.

Now assume that L is some logic and $\mathsf{LC} \nsubseteq L$; then $L \nVdash \mathcal{J}(\mathsf{2F})$ or $L \nVdash \mathcal{J}(\mathsf{Dia})$, so either the 2-fork or the diamond are frames for L. If $\mathsf{2F}$ is a frame for L, note that in Figure 1, we have the models M_1 and M_2, and $M_2 \leqslant_2 M_1$: if we pick in M_1 the top node labelled with 1, there is a node in M_2 with the same label, obtaining an isomorphism; and if we pick in M_1 the root, node that $M_1 \sim_1 M_2$. However, $M_2 \nleqslant_3 M_1$, since by picking the root of M_1, there is no 2-bisimilar node in M_2. Similarly, the models M_3 and M_4 (the frame of which is a p-morphic image of M_3) have the property that $M_3 \leqslant_2 M_4$ (there is an isomorphic node in M_3 for each node but the root, and the roots are 1-bisimilar), but $M_2 \nleqslant_3 M_4$. This shows that L cannot be 2-uniformly locally tabular again by Proposition 2.13. ∎

The case of 3-uniform logics is a fair bit more complicated and deserves further consideration. For that purpose, recall the logic wPL:

$$\mathsf{wPL} := \mathsf{IPC} \oplus (q \to p) \vee ((p \to q) \to p) \to p.$$

It is known (see [14, A.9]) that a Heyting algebra H validates wPL if and only if it is a *cascade Heyting algebra*. Their variety was named by Esakia CHL, and was studied in [6] in the context of analyzing profiniteness and representability of Heyting algebras. The finite members of such a variety are the "sums" of Boolean algebras; we will make use of the dual posets of such algebras:

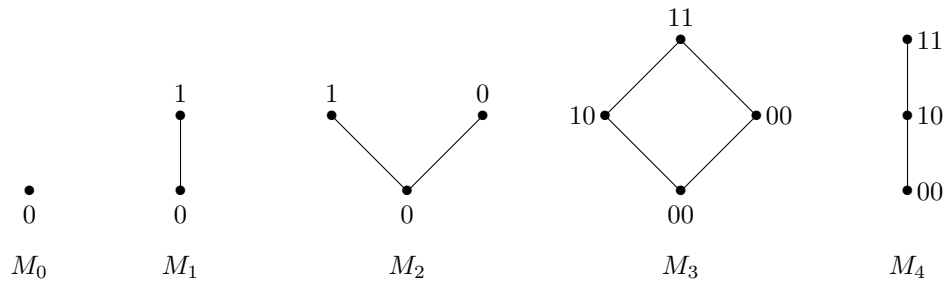

Figure 1

Definition 3.3. A finite (rooted) poset $(P, \leqslant)$ is called a *Boolean sum* if $d_p(P) = n$, and:

1. Whenever $x \prec y$, i.e., y is an immediate successor of x, then $d_p(x) = d_p(y) + 1$;
2. Each point x of depth $k+1$ sees all points of depth k for each k.

When working with Boolean sums, note that the notion of depth of a point captures precisely its *level*: for P a Boolean sum, $\mathsf{Lev}_k(P) = \{x \in P : d_p(x) = k\}$ consists of incomparable points which are all seen by the points in $\mathsf{Lev}_{k-1}(P)$, and all see the points in $\mathsf{Lev}_{k+1}(P)$. The following is a consequence of [14, A.9.2]

Proposition 3.4. A finite rooted poset P is a frame for wPL if and only it is a Boolean sum.

The following was shown by Mardaev [23, Theorem 5]:

Theorem 3.5. *The logic* wPL *can be axiomatized as:*

$$\mathsf{wPL} := \mathsf{IPC} \oplus \mathcal{J}(Q_1) \oplus \mathcal{J}(Q_2) \oplus \mathcal{J}(Q_3),$$

where $\mathcal{J}(Q_i)$ is the Yankov formula for the posets in Figure 2. Moreover, this logic is locally tabular.

We will be concerned with special extensions of this logic. For that we will need some definitions:

Definition 3.6. Let P be a Boolean sum of depth n. For each n, let c_n be the cardinality of the set of elements of depth n. Given $\langle c_1, \ldots, c_n \rangle$ we say that a consecutive subsequence $c_i, \ldots, c_{i+k}$ where $c_{i+j} > 1$ for $0 \leqslant j \leqslant k$ is a *k-stack*. We say that P has *stack-depth* k if there exists a k-stack but not a $k+1$-stack in P.

The stack depth of a Boolean sum thus measures how many consecutive levels there can be which contain more than one point.

Remark 3.1. In [6] a subvariety of CHL was considered in connection to the problem of profinite completions, called the variety of *Diamond Heyting algebras* DHL. Such algebras are of width 2, have a maximal element, and they satisfy the Yankov formula $\mathcal{J}(Q_5)$. Thus, their finite rooted posets are precisely Boolean sums of stack depth 1, width 2 having a greatest element (see also Section 4, where the associated logic will be called Box).

Lemma 3.7. Let (P, x) be a finite poset which is a Boolean sum. Then $P \Vdash \mathcal{J}(Q_i)$, for $i \in \{4, 5\}$ in Figure 2, if and only if P has stack-depth 1.

Proof. First note that both of the posets Q_4, Q_5 have stack depth 2. If Q_4, Q_5 is a rooted subframe of Q, which is a p-morphic image of P, by pulling along the p-morphism we will see that P has stack depth at least 2.

Conversely, assume that P has stack depth at least 2. Let $x \in P$ be a maximal point such that $\uparrow x$ has stack-depth 2. We may thus assume that x has at least two successors y_0, y_1, and that each of them has two

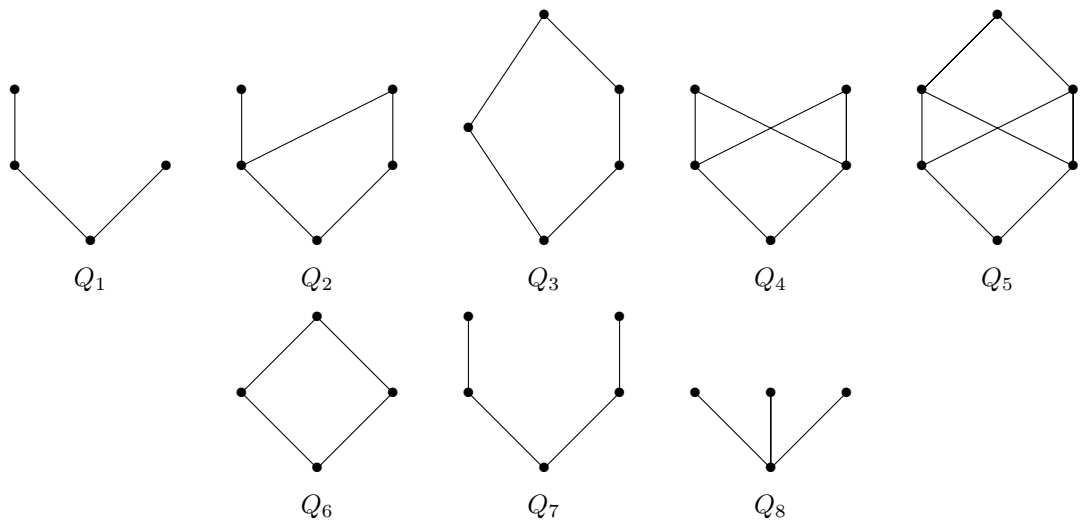

Figure 2: Q_i posets

successors (otherwise we could have picked y_i as our maximal point) z_0, z_1. Then there are two cases: either z_0 and z_1 have no common successor, or they do. In the first first case, we define a map to Q_4 as follows: we send z_0 to the left top point in Q_4, z_1 and all remaining points of that depth to the right top point; y_0 to the left middle point, y_1 and all remaining points of that depth to the middle right point; and x to itself. This is a surjective p-morphism onto Q_4. The case of Q_5 is similar, except we send all points above z_0, z_1 to the maximal point. ■

Given a rooted frame (P, x) based on a Boolean sum with stack depth 1, let us say that a point $y \in P$ is *solitary* if y is the only point of depth k for some k. Otherwise we say that y is *accompanied* by a point y'. The importance of stack depth for us comes from the following result:

Lemma 3.8. Let (P, x) be an n-generated model based on a finite Boolean sum of stack depth 1. If $(P, x) \sim_1 (P_y, y)$ for some $y \in \uparrow x$, then $x = y$.

Proof. Assume first that x is a solitary point. Then for each y, either $x \leqslant y$ or $y \leqslant x$ must hold. If $y \leqslant x$, note that by n-generation, any predecessor of x must have a color strictly smaller than x, and thus we cannot have $(P, y) \sim_1 (P, x)$. On the other hand, if $x \leqslant y$, let $S(x)$ be the set of immediate successors of x. If $S(x)$ is a singleton, we likewise have that the color of x must be strictly smaller, and so $(P, y) \sim_1 (P, x)$ cannot hold. If $S(x) = \{x_0, x_0'\}$, then since the stack depth is 1, these points must have distinct color, and either $c(x_0) \neq y$ or $c(x_0') \neq y$; say $c(x_0) \neq y$ without loss of generality. If $(P, y) \sim_1 (P, x)$, then y would have to have a successor $y \leqslant y_0$ such that $c(y_0) = c(x_0)$. However, note that if z is the unique immediate successor of x_0, x_0', $c(x_0), c(x_0') < c(z)$, so $y_0 = x_0$ or $y_0 = x_0'$. Thus if $x \neq y$, we would have a contradiction, since $x \leqslant y \leqslant y_0$ would force $y = y_0$ (since y_0 must be an immediate successor of x), whilst we chose $c(y) \neq c(y_0)$. It follows that $(P, x) \sim_1 (P, y)$ can only hold if $x = y$.

Next assume that x is accompanied by x'. If $x < y$, then by n-generation, $c(x) \neq c(y)$. Moreover, $c(x) \neq c(x')$, so $(P, x) \sim_1 (P, x')$ cannot hold. Finally if $y < x$, then by the same arguments as in the previous case, if $(P, x) \sim_1 (P, y)$, we would obtain a contradiction. We thus obtain the result. ■

Proposition 3.9. Let P, Q be two finite Boolean sum frames of stack depth 1. Assume that $(P, x) \sim_2 (Q, y)$ are n-generated models which are 2-bisimilar. Then they are isomorphic.

Proof. Let $S_2 \subseteq S_1 \subseteq S_0 \subseteq P \times Q$ be a 2-bisimulation. We claim that S_1 defines an isomorphism, mapping x to y. Indeed, assume that $y, y' \in P$ and $(y, z) \in S_1$ and $(y', z) \in S_1$. Then we have that $(P, y) \sim_1 (Q, z) \sim_1$

(P, y'), so $(P, y) \sim_1 (P, y')$. By Lemma 3.8, we have that $y = y'$. Thus S_1 is injective; a similar argument applied to Q shows that S_1 is functional, and the bisimulation clauses guarantee that it is a surjective relation. Thus S_1 is an isomorphism as desired. ■

Let 2Uni be the following logic:

$$\mathsf{2Uni} := \mathsf{IPC} \oplus \{\mathcal{J}(Q_i) : i \in \{1, \ldots, 5\}\} = \mathsf{wPL} \oplus \mathcal{J}(Q_4) \oplus \mathcal{J}(Q_5).$$

Then we have:

Theorem 3.10. *The logic* 2Uni *is the least* 3*-uniform logic.*

Proof. First, note that in Figure 3, we see that all the posets Q_i for $1 \leqslant i \leqslant 5$ admit 2-bisimilar models (obtained by p-morphic images of Q_i, denoted Q_i^l and Q_i^r whether they occur on the right or left) which violate our criterion. Namely, we have that $Q_i^l \leqslant_3 Q_i^r$, since for each node except the root in Q_i^r there is a point isomorphic to it in Q_i^l, and the root of Q_i^r is 2-bisimilar to Q_i^l; but $Q_i^l \nleqslant_4 Q_i^r$, because the root in Q_i^r has no 3-bisimilar point on the left models.

Hence, no logic which contains them can be 2-uniform. By our axiomatization of the logic 2Uni, any logic which is not an extension of it will therefore contain one of these posets by Yankov's theorem.

On the other hand, by Lemma 3.7, Proposition 3.9 and Corollary 2.13, we have that 2Uni is 3-uniform. ■

To finish this section we note that there will only be countably many logics which are 2-uniform, we recall that we can consider an order on finite rooted posets: $Q \trianglelefteq P$ if and only if Q is a p-morphic image of a rooted upset of P; this is known to form a partial order, which is well-founded, since the posets are finite. Moreover the following is well-known (see [4, Section 3]):

Theorem 3.11. *Let L be a locally tabular logic, and $(\mathbf{RP}(L), \trianglelefteq)$ the poset of rooted posets of L ordered by $\trianglelefteq$. If $(\mathbf{RP}(L), \trianglelefteq)$ contains no infinite antichains, then L has only countably many extensions, and all of them are finitely axiomatizable.*

Corollary 3.12. *There exist only countably many* 2*-uniform logics.*

Proof. Note that by Lemma 3.7, we have that any rooted poset P corresponds to a pair, of a finite sequence and a natural number, $((k_0, ..., k_{n-1}), k_n)$ where the sequence has k_0 always 1, and the numbers are alternatingly 1. We denote by $\mathsf{seq}(P)$ this tuple associated to each poset. We order such pairs as:

$$((k_0, ..., k_{n-1}), k_n) \preceq ((j_0, ..., j_{t-1}), j_t),$$

if and only if $(k_0, ..., k_{n-1})$ is a subsequence of $(j_0, ..., j_{t-1})$, and $k_n \leqslant j_t$. Then note that:

$$Q \trianglelefteq P \iff \mathsf{seq}(Q) \preceq \mathsf{seq}(P);$$

indeed, first note that if Q is a rooted upset of P', it is in fact a p-morphic image as well, since one can collapse all lower levels to the root of Q. Hence we have that $Q \trianglelefteq P$ if and only if there is a p-morphic image from P onto Q. Now if this happens, then what can happen to P is that the number of levels in P can get shorter, or the number of elements in the levels may get smaller – the p-morphism defines an inclusion of $\mathsf{seq}(Q)$ in $\mathsf{seq}(P)$, with the last element being preserved. The converse is likewise easy to see.

We thus obtain an isomorphism of $(\mathbf{RP}(L), \trianglelefteq)$ with $\omega^{<\omega} \times \omega$ ordered by the product of the subsequence order and the natural number ordering. By Higman's lemma, the set $\omega^{<\omega}$ ordered by subsequence is a well-partial order, and the product of well-partial orders is again one by Dickson's lemma. So $\mathbf{RP}(L)$ contains no infinite antichains, which by Theorem 3.11 means that there are only countably many extensions of 2Uni, all of which are finitely axiomatizable. ■

Remark 3.2. Corollary 3.12 should be contrasted with the fact that there are uncountably many logics of depth 3 (see [4, Lemma 3.19, Antichain Ω_2], and as noted by Shehtman, all logics of depth 3 are 5-uniform.

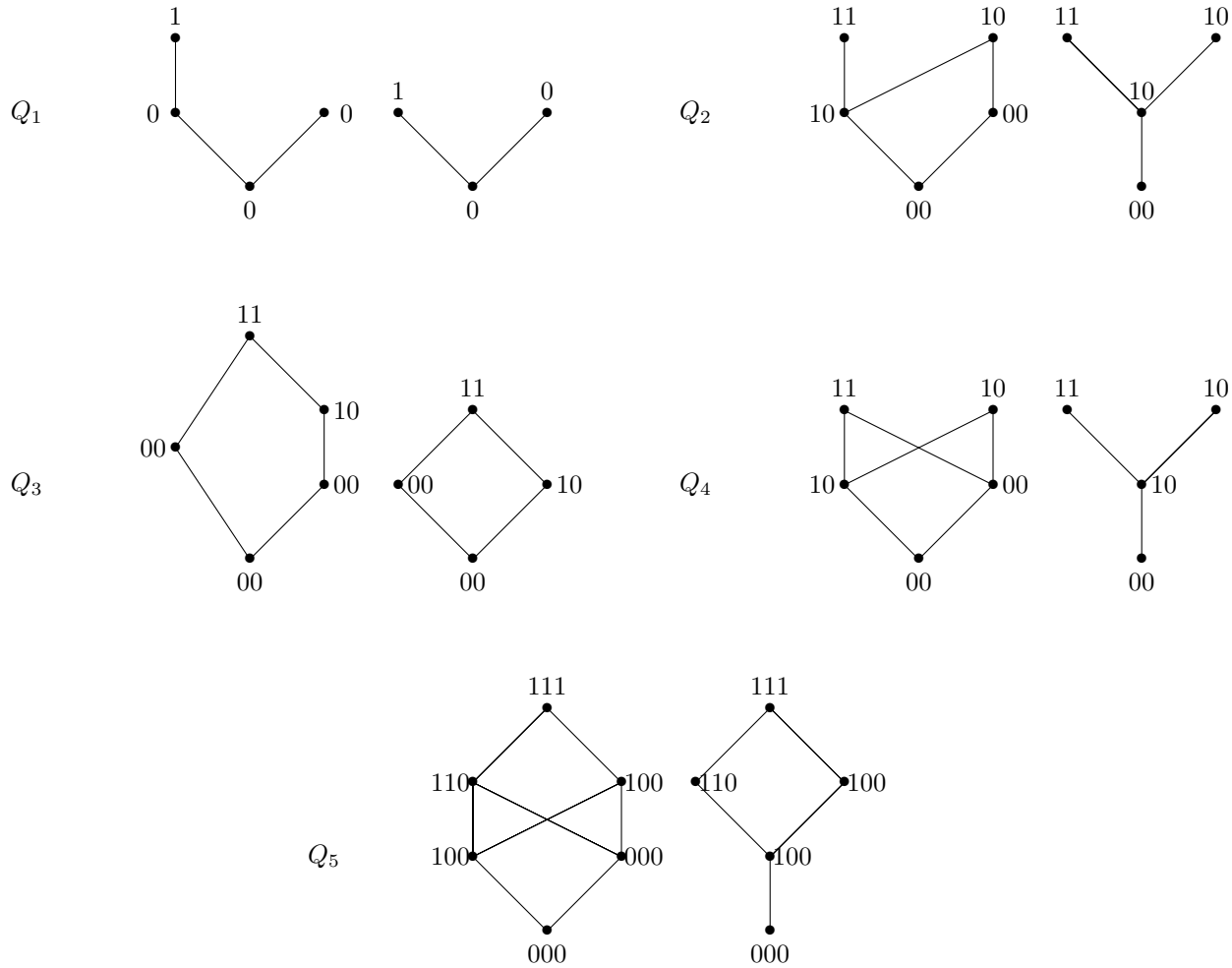

Figure 3: Models witnessing failure of 3-uniformity

3.2 3-uniformity and the logic of finite Combs

It is clear from the aforementioned discussion that 2-uniformity is substantially more complicated than 1-uniformity, and that it is far from exhausting all uniformly locally tabular logics. In this section we study a basic example of a 3-uniform logic of infinite depth, unrelated to wPL, and which might be interesting in its own right. The logic provided is the intuitionistic analogue of the bi-intuitionistic logic introduced by Martins and Moraschini [24], biLFC – the logic of finite combs.

Definition 3.13. A poset P is called an *n-comb* if $P = P_0 \sqcup P_1$, where $P_0 = \{x_1, \dots, x_n\}$ and $P_1 = \{y_1, \dots, y_n\}$, and we have that for each $x \in P$,

1. y_i is maximal;
2. $x_i \leqslant x$ if and only if $x = x_j$ and $i \leqslant j$, or $x = y_j$ and $i \leqslant j$.

A subposet $Q \subseteq P$ is called a *broken n-comb* if $P_0 \subseteq Q$. We call a poset a (broken) comb if it is a (broken) n-comb for some n.

The following is easy to verify:

Lemma 3.14. A poset is a p-morphic image of an n-comb if and only if it is a broken m-comb.

Proof. Simply note that for n-combs, the maximal points must be identified with each other. ■

Note that the posets in Figure 1 and in Figure 4 are all models over combs and broken combs. The poset Q_1 is likewise an example, and the depiction in Figure 3 shows that 2-bisimilarity is not enough to capture full bisimilarity for such models.

Proposition 3.15. Let P and Q be broken combs. Let (P, x) and (Q, x') be n-generated models which are 3-bisimilar. Then they are isomorphic.

Proof. Assume that the two posets are n-generated for some n. Now, if (P, x) and (Q, x') are 3-bisimilar, notice that they must have the same maximal elements. Suppose that (P, x) has a largest non-maximal element which has a single immediate successor, x_k; if (Q, y) did not end have a similar looking point, look at x'_k, its last point in the base of the comb, which will have two immediate successors. Then by the 3-bisimulation, there must be a point x_j which is 2-bisimilar with x'_k. Such a point must see two distinct maximal elements, and hence it must be some point $x_j < x_k$.

First assume that x_j has a different color from x_k; then this forces that some point of color x_k will be above x'_k, and by 3-bisimulation, we get a contradiction. So if x_j has the same color, we can assume that it sees exactly two maximal elements, and hence, is covered by x_k. By 3-bisimulation, there must be a point $x''_k \in Q$ which is 2-bisimilar with x_k – but by the configuration, such an element cannot exist. So we have that x'_k must have a single maximal element.

Now let x_{k-1} be the immediate predecessor of the maximal point in P. We show that Q must have a point with the exact same structure. If x_{k-1} has a single successor, x_k, the argument is similar to the one given in Proposition 3.9. If x_{k-1} has another immediate successor y_{k-1}, then we proceed as in the previous paragraph to ensure that Q has such a point in the immediate predecessor of x'_k. ■

Let LFC be the following logic:

$$\mathsf{LFC} := \mathrm{Log}(\{P : P \text{ is an } n\text{-comb}\}).$$

We have thus shown:

Theorem 3.16. *The logic* LFC *is uniformly locally tabular. Indeed it is* 4*-uniform, but not* 3*-uniform.*

Proof. By Corollary 2.13, and Proposition 3.15, we have the 4-uniformity of LFC. The fact that it is not 3-uniform follows from Theorem 3.10. ■

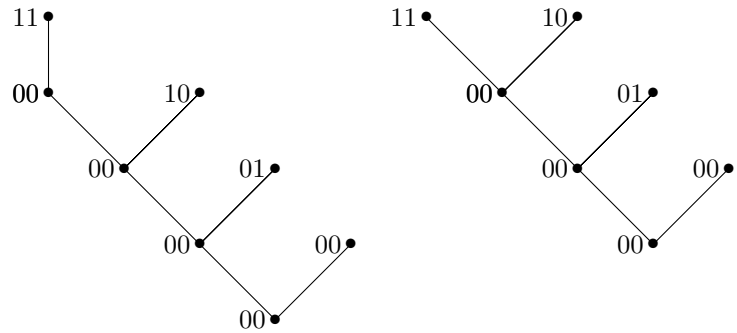

Figure 4: Comb and broken comb models

We conclude by noting that this logic admits a simple axiomatization with Yankov formulas. For that we need the following lemma:

Theorem 3.17. *The logic:*

$$\mathsf{IPC} \oplus \{\mathcal{J}(Q_i) : i \in \{2, 4, 5, 6, 7, 8\}\},$$

is locally finite, and its finite models are the finite combs. Consequently, it coincides with LFC.

Proof. For a given n we will describe the most general n-generated model possible for this logic[2]. Such a model will be shown to be finite.

Indeed, note that we can start with 2^n many elements. Now if x has one of the maximal elements as its cover, it can only have at most two of them, otherwise we obtain a copy of Q_8 as a rooted upset. Now let x be a point such that it is not covered by a maximal element; enumerate its successors, say $y_1, \ldots, y_n$. Note that if y_i and y_j never see the same maximal elements, then $n = 1$; otherwise we would obtain a p-morphism onto Q_7.

1. Assume that y_i and y_j both see the same maximal element z. Suppose that there is no other z'. Then we get a p-morphism onto Q_6.

2. Assume that there is at least another maximal element z'. Identify all other maximal elements with z'. Then either y_i and y_j both now see z', or one sees both z and z' and the other only sees z. Then we obtain a p-morphism either onto Q_2 or Q_4.

This shows that $n = 1$ is the only possibility. Hence if x does not see any maximal element, its color must be strictly smaller than that of its unique immediate successor. This shows that P must look like a broken k-comb for some k. As such, we get that this coincides with LFC, since it is not hard to show that every poset of LFC validates the formulas of the above logic. ■

3.3 A non-uniformly locally tabular logic

The analysis of 3-uniformity sheds some light on the possibilities for logics to be n-uniform but not $(n+1)$-uniform. Namely, note that $\mathsf{Log}(Q_4)$ will be an extension of wPL, and it will be 4-uniform but not 3-uniform by what was just shown. A more general fact holds which will allow us to provide an example of a locally tabular logic which is not uniformly locally tabular.

Definition 3.18. Let $n > 2$ and $k + 1 < n$. Let M_n^k, N_n^k for $k \leqslant n$ be the models depicted in Figure 6. Formally, we define:

[2]This is essentially a description of the so-called n-universal model for this logic; see [8, Chapter 3] for precise details on this construction.

1. M_n^k has $k+1$ many layers of two elements; if $L_j(M_n^k)$ is the j-th layer for $0 \leqslant j \leqslant k$, then the left element in $L_j(M_n^k)$ is labelled with $1^{n-j}0^j$, and the right element with $1^{n-(j+1)}0^{j+1}$. There is also a root element labelled with $1^{n-k-1}0^{k+1}$.

2. N_n^k has likewise $k+1$ many layers of two elements; if $L_j(N_n^k)$ is the j-th layer for $0 \leqslant j \leqslant k$, then the left element in $L_j(N_n^k)$ is labelled with $1^{n-j}0^j$, and the right element with $1^{n-(j+1)}0^{j+1}$. There is also a root element labelled with $1^{n-k-2}0^{k+2}$.

Thus the only difference between M_n^k and N_n^k is in the label of the root. The order in both cases is given by stacking the layers on top of each other.

Note that these models are always n-generated. Indeed, the key difference between these models is that the root of N_n^k goes down in color, relative to its right immediate successor, whilst in M_n^k the color is the same. Figure 5 shows the cases for $k = 1$, for any $n \geqslant 2$

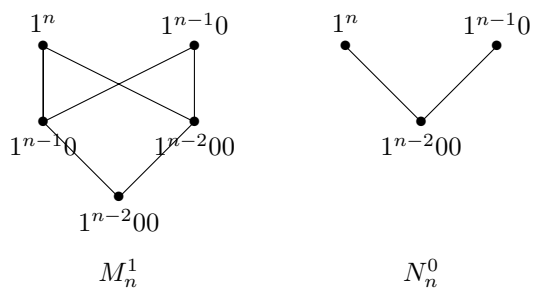

Figure 5

The following lemma provides the relationship between such models:

Lemma 3.19. Let $n > 2$ be arbitrary, and $k+1 < n$. Then M_n^k is k-bisimilar, but not $(k+1)$-bisimilar with N_n^{k-1}.

Proof. This is shown by induction. Note that for $k = 1$, this is easy to verify, since N_n^1 will be of depth 2: looking at Figure 5, we have that the two roots have the same color, and they see points with the same colors, so they will be 1-bisimilar. However, assume that they were 2-bisimilar; let x be the point in M_n^1 with the label $1^{n-1}0$ which is the successor of the root. Then by 2-bisimilarity, there is some point in N_n^0 which is 1-bisimilar with it; but the only point which could be is the maximal point, which is a contradiction.

Now, assume that this holds for k. Consider M_n^{k+1} and N_n^k. Note that they have roots of the same color, labelled with $1^{n-k-2}0^{k+2}$, and indeed, if z is the immediate successor of M_n^{k+1} with the same color, $\uparrow z$ is isomorphic to N_n^k. Thus if Player I picks the root of N_n^k or any of its successors, Player II has an obvious response by choosing the element inside of $\uparrow z$. Thus the only way that Player I can hope to win the game is to pick either the root of M_n^k, or the left successor of the root. If Player I picks the left successor of the root, say y_0, note that $\uparrow y_0 \cong M_n^k$; on the other hand, the right successor of the root of N_n^k is isomorphic to N_n^{k-1}, so Player II wins by induction hypothesis. Finally, if Player I picks the root of M_n^{k+1}, Player II is forced to show that this is k-bisimilar to the root of N_n^k; by the arguments just shown Player I is in turn forced to play the root of M_n^{k+1} for all k many steps of the game, and loses when Player II finally picks the root, given the two roots have the same color. This shows that M_n^{k+1} is k-bisimilar to N_n^k as desired.

Now assume that M_n^{k+1} is $(k+1)$-bisimilar to N_n^k. Then Player I can pick the left successor of the root of M_n^{k+1}, y_0, labelled with $1^{n-k-1}0^{k+1}$, and Player II is forced to respond with the right successor of N_n^k, y_1; as noted $\uparrow y_0 \cong M_n^k$ and $\uparrow y_1 \cong N_n^{k-1}$. Since by induction hypothesis these models are not k-bisimilar, Player II then loses, meaning that M_n^{k+1} and N_n^k are not $(k+1)$-bisimilar. ■

Hence we have the following:

Theorem 3.20. *The logic* wPL *is locally tabular but not uniformly locally tabular.*

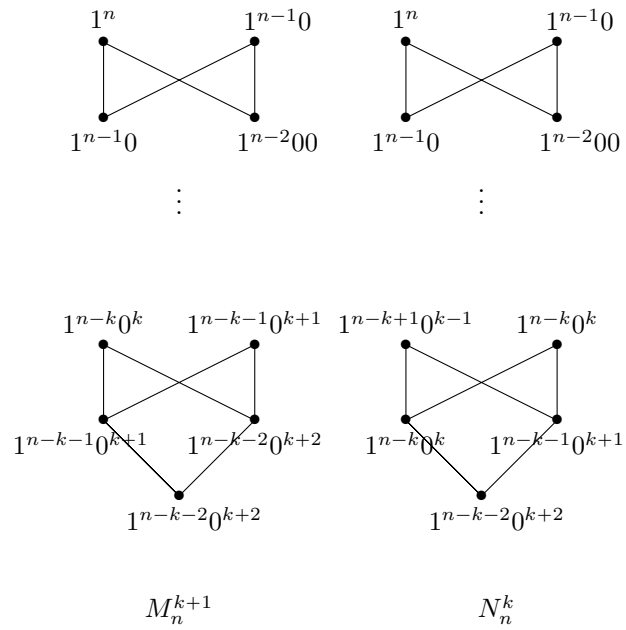

Figure 6: General models M_n^{k+1} and N_n^k

Proof. All the models M_n^k for each n, k are based on frames which validate wPL; hence by Corollary 2.13, wPL is not uniformly locally tabular, since we notice that $M_n^{k+1} \leqslant_{k+1} N_n^k$ but not $M_n^{k+1} \leqslant_{k+2} N_n^k$. On the other hand, it is locally tabular by Theorem 3.5. ■

4 Pre-uniform local tabularity

As noted before the problem of working with locally tabular intermediate logics presents a challenge since, as shown by Mardaev [22], there are uncountably many such pre-locally tabular logics. By contrast, there are exactly three pretabular logics [15], and every non-tabular logic is contained in one of them, and all of them are decidable. Together, this makes the problem of deciding, given an axiom φ, whether $\mathsf{IPC} \oplus \varphi$ is tabular or not decidable. We thus have the following interesting problem:

Problem 1. *Is uniform local tabularity decidable over* IPC*?*

Intrinsic in this is the idea of "pre-local tabularity". The idea of "pre-P"-logics, for P a property, has been extensively used by several authors (e.g., [15, 25, 20]) to characterize certain logical properties. We can make such a definition also in our setting:

Definition 4.1. Let L be a superintuitionistic logic. We say that L is *pre-uniformly locally tabular* if L is not uniformly locally tabular but all of its extensions are so.

We will now characterize pre-uniform local tabularity amongst extensions of KG. For that, first recall the following well-known formula,

$$\mathsf{bw}_2 \coloneqq (p_0 \to (p_1 \vee p_2)) \vee (p_1 \to (p_0 \vee p_2)) \vee (p_2 \to (p_0 \vee p_1)),$$

the *bounded width 2* formula [12, Proposition 2.39]; a finite frame validates this axiom if and only if they have width at most 2, i.e., for each point x in such a frame, $\uparrow x$ contains antichains of size at most 2. Recall also the formula:

$$\neg p \vee \neg\neg p,$$

the weak law of excluded middle; it is also well-known [12, Proposition 2.37] that frames for such a formula are exactly those containing a greatest point above each point.

Definition 4.2. The logic Box is defined to be the extension of wPL by the $\mathsf{Box} = \mathsf{wPL} \oplus \mathsf{bw}_2 \oplus (\neg p \vee \neg\neg p)$.

We will see that rooted frames for Box look very much like those from Figure 6, with a point in the maximum – if one thinks of four adjacent points there, these look like stacks of boxes.

Remark 4.1. Note that Box, as an extension of wPL is locally tabular. For that reason, we will assume throughout that all logics L extending it are locally tabular, and consequently have the finite model property.

We will now show that Box is pre-uniformly locally tabular. For that purpose, let S_n be the result of adding a maximal point to the underlying frame of N_n^{n-2} (see Figure 6). We denote by $\mathsf{Lev}_j(S_k) = \{x \in S_k : d_p(x) = j\}$, the *j-th level*. Then first we show:

Proposition 4.3. We have that $\mathsf{Box} = \mathsf{wPL} \oplus \mathsf{bw}_2 \oplus (\neg p \vee \neg\neg p) = \mathrm{Log}(\{S_n : n \in \omega\})$.

Proof. It is clear that each S_n validates all the axioms of Box, since they have a maximal point, have width at most 2, and are frames for wPL as noted in Theorem 3.20. Conversely, by the semantic characterizations given of these formulas, any finite rooted poset P validating Box will be a Boolean sum (as it validates wPL), it will have width at most 2 by bw_2, and a maximal element by the axiom $\neg p \vee \neg\neg p$. Let k be the depth of the root of P. Then for any $j \leqslant k$, either there is a unique point x such that $d_p(x) = j$, or there are two. This defines a p-morphism from S_k onto P. Thus, every finite rooted poset of Box can be obtained by p-morphisms from one of the form S_n, showing the result. ■

Recall the notion of stack depth (Definition 3.6); for frames with bounded width 2, having stack depth k amounts to having a bound of k on the number of consecutive levels with two points. Thus we obtain:

Proposition 4.4. Let $\mathsf{Box} \subsetneq L'$ be a logic of infinite depth. Then there is some n such that posets in L' have stack depth n.

Proof. If we assume that there is no bound on the stack depth of posets for L', then for each n, let P be a frame for L' containing a stack of at least $n+2$ many elements. Let p be one of the minimal elements of the stack, and m be an element situated at at n many levels above p; then $\uparrow p$ is a frame for L'. Define a p-morphism onto S_n by sending m, the other element in its level, and all their successors to the top element in S_n, and all elements in the stack mapped through an isomorphism to S_n. Thus $S_n \Vdash L'$. Since this holds for any n, by Proposition 4.3 we have that $L' = \mathsf{Box}$. ■

We thus obtain that the extensions of Box are controlled by the depth, which can be infinite or finite, and the stack-depth, which can be finite or infinite. We show the following:

Proposition 4.5. Assume that $L \supseteq \mathsf{Box}$ is a logic of infinite depth such that all its elements have at most stack-depth $k \geqslant 1$. Then L is $(k+2)$-uniformly tabular. Consequently, Box is pre-uniformly locally tabular.

Proof. The fact that Box is not uniformly locally tabular follows by a simple modification of Theorem 3.20, adding a point on top to the models M_n^k with a distinguished color.

Assume that $L \supseteq \mathsf{Box}$ is a logic such that its models have stack depth at most k. Then we show the following modification of Lemma 3.8: if (P, x) is a Boolean sum of stack depth at most k and $(P, x) \sim_k (P_y, y)$ then $x = y$. This proceeds by a double induction on k and m, where m is the number of stacks above x of size at least k. For $k = 1$ and arbitrary m, this was provided by Lemma 3.8. Assume that it holds for k, and we first show it for $k+1$ and $m = 1$; In this case, note that since $(P, x) \sim_{k+1} (P_y, y)$, we can pick an immediate successor of x, x', and by k-bisimilarity, $(P_{x'}, x') \sim_k (P_{y'}, y')$ for some sucessor of y; then the stack above $P_{x'}$ must be of size at most k, so by induction hypothesis, $x' = y'$ must hold. Since this holds for any immediate successor of x, we conclude that $x = y$. Assuming this for arbitrary m, we then show it for $m+1$, by noting that when we move to an immediate successor, the immediate successor will see one fewer $k+1$-stack, and so the induction hypothesis can be carried through.

From this fact, we proceed as in Proposition 3.9: assuming we have $(P, x), (Q, y)$, n-generated models based on frames for L, where L has a bound on the stack-depth of k, such that $(P, x) \sim_{k+1} (Q, y)$. Then we have an S_{k+1}-bisimulation, defined by a chain $S_{k+1} \subseteq S_k \subseteq \cdots \subseteq S_0$; then the relation S_k defines an isomorphism, and so (P, x) and (Q, y) are bisimilar. From this it follows by the same arguments as in Proposition 3.9 that L is $k + 2$-uniformly locally tabular. ■

4.1 Pre-uniform local tabularity in KG

We will provide a solution to Problem 1 in the case of extensions of the system KG of Kuznetsov-Gerciu [1] (for an overview of the system see [8, Chapter 4]). Such a system has a very broad range of extensions, and a rich semantics, which has made it useful for several purposes in the study of superintuitionistic logic (e.g. [5]). One of its key extensions is the logic $\mathsf{RN} = \mathrm{Log}(\mathsf{RN})$ – the logic of the frame of the Rieger-Nishimura ladder (see Figure 7), which is dual to the *Rieger-Nishimura lattice*, the free 1-generated Heyting algebra.

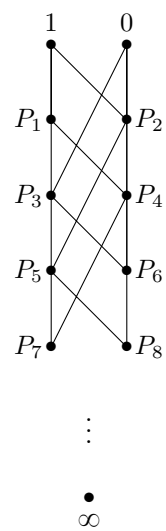

Figure 7: Rieger-Nishimura Ladder RN

Recall from [16, 10] (see also [9] and [2]) that RN, the Rieger-Nishimura ladder, can be constructed as the inverse limit of a sequence:

$$P_1 \xleftarrow{r_1} P_2 \xleftarrow{r_2} P_3 \xleftarrow{r_3} \dots$$

where r_i is a map which is a p-morphism with respect to "sufficiently many" parts of P_3. In fact these admit an explicit description:

1. The posets P_i are precisely the upsets as labelled in Figure 7.
2. The map $r_{2n+1} : P_{2n+2} \to P_{2n+1}$ sends the P_{2n+2}-root point, and the P_{2n}-root point, to P_{2n+1}, and leaves the rest untouched;
3. The map r_{2n+2} sends the P_{2n+3} and the P_{2n+1}-root points to P_{2n+2}, and leaves everything else untouched.

For our purposes we will need the following lemma, shown essentially in [8, Chapter 4]:

Lemma 4.6. For each n, the poset S_n is an upset of a p-morphic image of $P_{2n+1} \oplus 1$.

Proof. One proceeds inductively as follows: first identify the point labelled 1 in Figure 7 with the top element. This produces one top level containing the maximum, followed by a level containing P_1 and 0, seen by P_3. Then to obtain the next layer, identify P_2 with its now unique successor, and we obtain a new layer, consisting of P_3 and P_4, seen by P_6. We thus proceed in this way recursively, obtaining all boxes as desired. ■

To characterize uniform local tabularity, we will rely on the fact that already local tabularity above KG is characterized. In principle this is not necessary, as uniform local tabularity is potentially simpler; however in this simple case, it will be convenient to rely on the following, shown in [8, Theorem 4.6.5]:

Theorem 4.7. *A logic* $\mathsf{KG} \subseteq L$ *is not locally tabular if and only if* $L \subseteq \mathsf{RN.KC} = \mathsf{RN} \oplus (\neg p \vee \neg\neg p)$.

We will also need one more technical notion:

Definition 4.8. We say that a poset P has *degree of uniformity* k if whenever (P, x) is an n-generated model, and (Q, y) is another n-generated model such that $S_k \subseteq P \times Q$, is a k-bisimulation between x and y, then S_k is an isomorphism.

Remark 4.2. The reader should notice that for a fixed poset P, a degree of uniformity need not be minimal – for the purposes where we will use it, the notion of a minimal degree of uniformity will not be needed.

It is well-known that any finite poset has a degree of uniformity (see e.g. [17, Proposition 4.5]). Recall that given rooted posets P, Q, we write $P \oplus Q$ for the *vertical sum* (see [8, Def. 4.1.13]) the poset based on the disjoint union of P, Q, where for each $x \in P$ and $y \in Q$, $x \leqslant y$. We denote the repeated vertical sum as $P_1 \oplus \cdots \oplus P_n$.

Lemma 4.9. If $P_1, \ldots, P_n$ is a sequence of posets with degree of uniformity k_i, then $P_1 \oplus \cdots \oplus P_n$ has degree of uniformity $k' = \max(\{k_i : i \leqslant n\})) + 1$.

Proof. Write $P = P_1 \oplus \cdots \oplus P_n$. Let (Q, y) be an arbitrary n-generated model, and assume that $(P, x) \sim_{k'} (Q, y)$. Let x_n be the root of P_n; by k'-bisimilarity, we have that there is some $y_n \in Q$, such that $(P_{x_n}, x_n) \sim_{k'-1} (Q_{y_n}, y_n)$; since $k_n \leqslant k' - 1$, this relation defines an isomorphism. Note that any predecessor of x_n or y_n must have a properly smaller color. Let $Q_n = \uparrow y_n$. By restricting the relation $S_{k'}$ to $P \backslash (P_n \backslash \{x_n\})$ and $Q \backslash (Q_n \backslash \{y_n\})$, we claim that we still have a k'-bisimulation: for each $x' \geqslant x$, such that $x' \in P \backslash (P_n \backslash \{x_n\})$, there must be $y' \geqslant y$ which is $k' - 1$-bisimilar, and this must lie also in $Q \backslash (Q_n \backslash \{y_n\})$. Thus we can proceed by induction, showing that each such restriction of S_k is an isomorphism. ■

We can thus conclude:

Theorem 4.10. *If* $L \supseteq \mathsf{KG}$ *is any logic, then* L *is not pre-uniformly locally tabular if and only if* $L \subseteq \mathsf{Box}$.

Proof. If $\mathsf{KG} \subseteq L$, and L is not locally tabular, we thus have by Theorem 4.7 that $L \subseteq \mathsf{RN} \subseteq \mathsf{Box}$. Hence assume that L is locally tabular but not uniformly locally tabular. We can assume by Theorem 2.12 that L is not of finite depth. Since L is locally tabular, there must be an n which is maximal such that $\uparrow P'_n$ is a frame for the logic. Hence by [8, Corollary 4.3.9], the finitely generated frames for L must be finite vertical sums of p-morphic images of upsets of P_n for $k \leqslant n$ (not necessarily rooted).

Now as noted in Lemma 4.6, for large enough P_n we can obtain S_n, and for large enough vertical sums, we can obtain all the S_n. Suppose that there is some n such that we cannot obtain S_n as a frame for L. Then there must be a bound on the size of the posets Q with a root and a maximal element which appear in vertical sums in L (otherwise using the technique of Lemma 4.6 we could obtain arbitrarily large S_n). Using Lemma 4.9 we can conclude that posets for L will have a maximal degree of uniformity k for some k – which implies (and is stronger than the fact that) whenever $(P, x) \leqslant_k (Q, y)$ for (P, x) and (Q, y) two models based on frames for L, then $(P, x) \leqslant_\infty (Q, y)$. By Proposition 2.13 we obtain that L is uniformly locally tabular – a contradiction.

So we must assume that L contains arbitrarily large vertical sums of P_n as frames. Thus as noted, we can obtain S_n for each n. By Proposition 4.3 we have that $L \subseteq \mathsf{Box}$ as desired. ■

Corollary 4.11. *Above* KG*, uniform local tabularity is decidable: if* $L = \mathsf{KG} \oplus \varphi$*, it is decidable whether* L *is uniformly locally tabular.*

Proof. Since the logic Box is finitely axiomatized and locally finite, it is decidable. Given L a decidable logic, one simply checks whether $L \subseteq \mathsf{Box}$, by checking whether Box contains the axiom φ, which is enough by Theorem 4.10. ■

5 Conclusions and future work

In this paper we provide a first systematic analysis of the concept of uniform local tabularity. Whilst such a concept coincides with local tabularity in the extensions of S4, we have shown, confirming a conjecture of Shehtman, that the two concepts come apart in IPC. By stratifying the concept, we analyse the case of n-uniform local tabularity for $n \in \omega$, showing that there is always the largest n-uniform logic in this sense; for $n \leqslant 3$, we give full axiomatizations, and provide an example illustrating the added complexities for $n = 4$. We also show that the logic Box, whose finite subdirectly irreducible models are exactly the finite projective Heyting algebras with a maximal element, is the unique pre-uniformly locally tabular logic in logics above the system KG. A visual summary of these results in terms of the hierarchy of superintuitionistic logics is given in Figure 8.

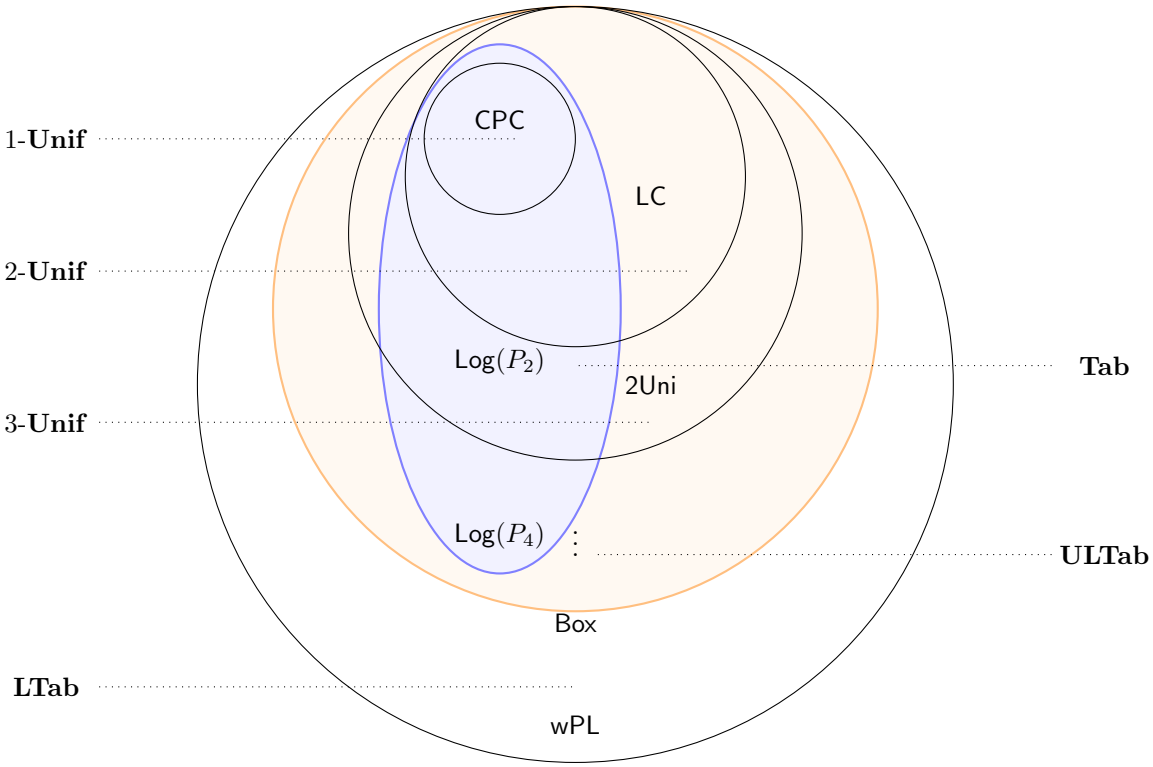

Figure 8: Hierarchy of Locally Tabular Superintuitionistic Logics; blue oval covers the tabular logics.

The present paper leaves many questions open. The following two are the most clear from our work:

1. Axiomatize the n-uniform logics for $n > 2$; the example of LFC can serve for this as a guiding light. It would be interesting in this respect to study the relationship between this and Maksimova's ω-operation studied in [21];

2. Analyze pre-uniform local tabularity above logics larger than KG – for example, BW_2, the logic of bounded width 2 – and find out whether it is decidable.

As a final note, we remark on the following point which was alluded to in the introduction, but certainly deserves further attention. In light of the results from [2], the property of n-uniformity can be stated as the appropriately relativised Ghilardi complex over a distributive lattice terminating after n many steps. This can be seen as meaning that the coproducts of the associated variety being "simple". This is explicit in the work of Carai [11], where an elegant description of coproducts of Gödel algebras is provided. It would thus be interesting to parallel the study of uniform local tabularity with the study of coproduct representations for uniformly locally finite varieties of Heyting algebras, and develop this in a systematic manner. We hope that the hierarchy of uniformly locally tabular logics proposed in this paper can serve as a basis for such a project.

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